Math Bootcamp

Day 5 — Dynamic Programming (Basics)

Math Camp 2026

Day 5 roadmap

1. **5.1** Sequential decisions and the DP idea
2. **5.2** State, control, and law of motion
3. **5.3** Two-period consumption–savings model
4. **5.4** Backward induction
5. **5.5** Bellman equation and policy function
6. **5.6** Euler equation and link to standard optimization

Focus today: one worked **two-period** example (finite horizon).

5.1 Why dynamic programming?

Many economic problems are **sequential**:

- You choose something **today**.
- It changes what you **can** choose tomorrow.
- Today's choice trades off **now** vs. **later**.

Examples. Consumption vs. savings; hiring vs. waiting; extracting a resource vs. leaving it in the ground.

Dynamic programming (DP). Break a multi-period problem into **smaller subproblems**, linked by a **value function**.

5.1 Finite horizon and the Markov idea

Finite horizon. Time $t = 1, 2, \ldots, T$ (today: $T = 2$). After period T, the problem ends (**terminal condition**).

Markov structure (informal). For decision-making, **only the current state matters —** not the full history, once the state is given. Example: tomorrow's opportunities depend on **assets on hand**, not on every past consumption choice separately.

Key objects. State (where you are), **control** (what you choose), **transition** (how state moves).

5.2 State versus control

Object	Meaning
State x_t	Variables summarizing the situation at the **start** of period t
Control u_t	Variables chosen **in** period t
Transition	Law of motion $x_{t+1} = g(x_t, u_t)$

Feasibility. Controls must satisfy constraints each period (budget, nonnegativity, etc.).

Payoff. Period-t utility $u(c_t)$ (or profit, welfare, ...). Future payoffs are discounted by $\beta \in (0, 1]$.

5.3 Two-period consumption–savings

Story. A household lives for **two periods**, $t = 1, 2$. Income $y > 0$ in period 1 only; no labor income in period 2. It chooses consumption $c_1, c_2 \geq 0$ and saves

$$s = y - c_1 \qquad \text{(savings in period 1).}$$

Savings earn gross return $1 + r$ ($r > -1$). Period-2 consumption is

$$c_2 = (1 + r)s = (1 + r)(y - c_1).$$

Preferences.

$$\max_{c_1, c_2} \log(c_1) + \beta \log(c_2), \qquad \beta \in (0, 1).$$

Log utility keeps algebra clean; same logic works for $u(\cdot)$ more generally.

5.3 State and control in this model

State at start of period t: assets b_t (wealth before choosing c_t).

- **Period 1:** $b_1 = 0$. Income y arrives; choose c_1. Assets carried to period 2:

$$b_2 = (1+r)(b_1 + y - c_1) = (1+r)(y - c_1).$$

- **Period 2:** state is b_2. Choose c_2 subject to $0 \leq c_2 \leq b_2$ (consume all assets by the end).

Controls: $u_1 = c_1$, $u_2 = c_2$. **Transition:** $b_{t+1} = (1+r)(b_t + y_t - c_t)$ with $y_2 = 0$.

5.4 Backward induction — idea

Strategy. In a **finite-horizon** problem, solve from the **last period backward**:

1. Period 2: optimize taking the state b_2 as **given**.
2. Period 1: optimize knowing **optimal future behavior** is already solved.

Why backward? In period 2 there is **no future left** — only current utility matters. Period 1 must account for how today's savings affect tomorrow's feasible consumption.

5.4 Step 1 — last period ($t = 2$)

Value function in period 2 (state b_2):

$$V_2(b_2) = \max_{0 \leq c_2 \leq b_2} \log(c_2).$$

Solution. Strictly increasing $\log(\cdot) \Rightarrow$ consume everything:

$$c_2^*(b_2) = b_2, \qquad V_2(b_2) = \log(b_2).$$

Policy function $c_2^*(b_2)$: maps state $\rightarrow$ optimal control. Here: "use all assets."

5.4 Step 2 — period $t = 1$

State: $b_1 = 0$, income y. Choosing c_1 sets $b_2 = (1 + r)(y - c_1)$.

Problem:

$$V_1(0) = \max_{0<c_1<y} \left\{ \log(c_1) + \beta V_2\big((1 + r)(y - c_1)\big) \right\}.$$

Substitute $V_2(b_2) = \log(b_2)$:

$$V_1(0) = \max_{0<c_1<y} \left\{ \log(c_1) + \beta \log\big((1 + r)(y - c_1)\big) \right\}.$$

FOC ($u = \log$):

$$\frac{1}{c_1} = \frac{\beta}{y - c_1} \quad \Longrightarrow \quad c_1^* = \frac{y}{1 + \beta}.$$

Then $c_2^* = (1 + r)(y - c_1^*) = \dfrac{\beta(1 + r)y}{1 + \beta}$.

5.5 Bellman equation — general form

Principle of optimality. Optimal plan $\Rightarrow$ remaining decisions are optimal from whatever state remains.

Recursive value function (finite T):

$$V_T(x_T) = \max_{u_T \in \mathcal{F}_T(x_T)} u(x_T, u_T)$$

$$V_t(x_t) = \max_{u_t \in \mathcal{F}_t(x_t)} \Big\{ u(x_t, u_t) + \beta V_{t+1}\big(g(x_t, u_t)\big) \Big\}, \quad t = 1, \dots, T-1.$$

Read the recursion. Current payoff + discounted **continuation value** V_{t+1} from next state.

5.5 Bellman equation — two-period special case

With $T = 2$, $b_1 = 0$, and $V_2(b_2) = \log(b_2)$:

$$V_1(0) = \max_{c_1} \left\{ \log(c_1) + \beta V_2\big((1+r)(y - c_1)\big) \right\}.$$

Policy functions:

$$c_1^*(0) = \frac{y}{1+\beta}, \qquad c_2^*(b_2) = b_2.$$

Value at start:

$$V_1(0) = \log\Big(\frac{y}{1+\beta}\Big) + \beta \log\Big(\frac{\beta(1+r)y}{1+\beta}\Big).$$

5.6 Euler equation (two periods)

Same optimum, different language. The FOC from backward induction is the **two-period Euler equation**:

$$u'(c_1^*) = \beta(1+r)\, u'(c_2^*).$$

For log utility, $u'(c) = 1/c$, so

$$\frac{1}{c_1^*} = \beta(1+r)\,\frac{1}{c_2^*} \quad \Longleftrightarrow \quad c_2^* = \beta(1+r)\, c_1^*.$$

Meaning. Marginal utility today equals discounted marginal utility tomorrow, times the return to saving. If $r = 0$: $c_2^* = \beta c_1^*$ (patient households consume more later only if $\beta < 1$ is balanced by returns).

5.6 Same answer from the lifetime budget constraint

Combine constraints ($b_1 = 0$):

$$c_1 + \frac{c_2}{1+r} = y.$$

Lagrangian:

$$\mathcal{L} = \log(c_1) + \beta \log(c_2) + \lambda\Big(y - c_1 - \frac{c_2}{1+r}\Big).$$

FOC give $1/c_1 = \lambda$ and $\beta/c_2 = \lambda/(1+r)$, hence $c_2^* = \beta(1+r)c_1^*$. Substitute into the budget:

$$c_1^*\Big(1+\beta\Big) = y \quad \Longrightarrow \quad c_1^* = \frac{y}{1+\beta}.$$

Takeaway. DP (backward) and static intertemporal optimization (one budget) are **two views of the same problem**.

5.6 Beyond two periods for interest only

for interest only

More periods. Same backward logic: solve V_T, then V_{T-1}, $\ldots$, down to V_1.

Infinite horizon. Value function iteration: start with a guess V^0, apply the Bellman operator until convergence. (Algorithm details: fall macro/micro courses.)

State may grow. With capital $k_{t+1} = (1-\delta)k_t + i_t$, the state is k_t; controls are c_t, i_t. Same template: $V_t(k_t) = \max\{u(c_t) + \beta V_{t+1}(k_{t+1})\}$.

Exercise. Same setup as above ($b_1 = 0$, income y in period 1, return $1 + r$), but $u(c) = \dfrac{c^{1-\sigma}}{1-\sigma}$ with $\sigma > 0$, $\sigma \neq 1$. Using backward induction, show

$$c_1^* = \frac{y}{1 + \beta^{1/\sigma}(1+r)^{(1-\sigma)/\sigma}}, \qquad c_2^* = \beta^{1/\sigma}(1+r)^{1/\sigma}\, c_1^*.$$

Solution (sketch). $V_2(b_2) = \max_{c_2 \le b_2} u(c_2) = u(b_2)$, so $c_2^*(b_2) = b_2$. Period 1 FOC:

$$c_1^{-\sigma} = \beta(1+r)^{1-\sigma}(y - c_1)^{-\sigma} \quad \Longrightarrow \quad \frac{c_2^*}{c_1^*} = \beta^{1/\sigma}(1+r)^{1/\sigma}.$$

Plug into $c_1 + c_2/(1+r) = y$ to get c_1^*. Check: $\sigma = 1$ (log) gives $c_1^* = y/(1+\beta)$.

5.6 Exercise: Income in both periods Take home

Exercise. Income $y_1, y_2 > 0$ in periods 1 and 2; gross return $1 + r$; utility $\log(c_1) + \beta \log(c_2)$. State $b_1 = 0$; transition $b_2 = (1 + r)(y_1 - c_1)$; period-2 budget $c_2 \le b_2 + y_2$. Derive $V_2(b_2)$, the Bellman equation for $V_1(0)$, and the optimal c_1^*, c_2^*.

Solution. $V_2(b_2) = \max_{0 < c_2 \le b_2 + y_2} \log(c_2) \Rightarrow c_2^*(b_2) = b_2 + y_2,\ V_2(b_2) = \log(b_2 + y_2)$.

$$V_1(0) = \max_{c_1} \left\{ \log(c_1) + \beta \log\big(y_2 + (1 + r)(y_1 - c_1)\big) \right\}.$$

FOC: $1/c_1 = \beta(1 + r)/(y_2 + (1 + r)(y_1 - c_1))$. Lifetime budget:
$c_1 + c_2/(1 + r) = y_1 + y_2/(1 + r)$ gives the same Euler equation $c_2^* = \beta(1 + r)c_1^*$.